Software Engineering Lab 6 & 7

Elevator Dispatch | Ramneek Gautam | PES1UG24AM220

LAB 6 - CODE AND BRANCH COVERAGE

The elevator dispatch module implements next_direction, door_action and pick_elevator. The test suite checks travel direction while moving up, down and idle; emergency, overload, blocked-door and safe-close decisions; and car selection including an empty car list, nearest car, moving-car penalty and tie handling.

Local result: 19 pytest tests passed.

LAB 7 - FUZZ TESTING AND INPUT VALIDATION

The functions parse_request, total_travel and load_percent process elevator requests, floor sequences and car load. Input validation covers malformed separators, missing values, extra fields, non-integer values, empty CSV entries, non-string inputs, negative loads and non-positive capacity.

The Hypothesis test suite defines generated text, integer-pair and integer-list properties. Hypothesis could not be installed in this execution environment because network access to package indexes was unavailable. Install requirements.txt and run python -m pytest test_lab7_processor.py -q for the actual Hypothesis run.

SAMPLE APPLICATION RUN

```
==== Running parse_request ====
Enter a request as FROM->TO (e.g. 5->9): From and To: (5, 9)

==== Running total_travel ====
Enter floors visited as CSV (e.g. 1,5,3,9): Floors Travelled: 12

==== Running load_percent ====
Enter load as 'X of Y' (e.g. 450 of 630): Load Percent: 71
```

Test Evidence

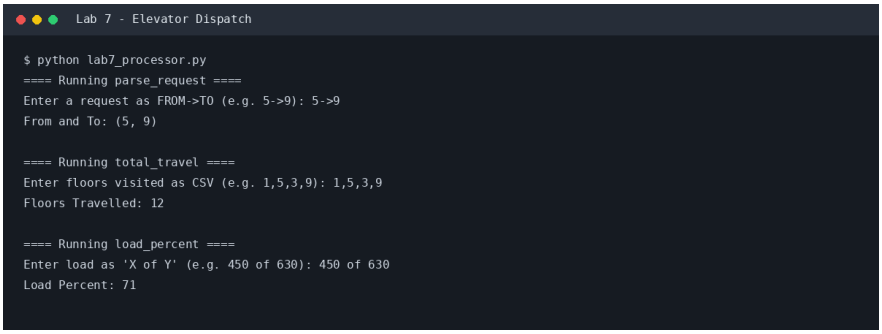
Local execution results

Lab 6: pytest run, 19 passed

Lab 6 - pytest

```
$ python -m pytest test_lab6_elevator_dispatch.py -q
...................                                      [100%]
19 passed in 0.07s
```

Lab 7: application sample output



Note: A genuine Hypothesis test screenshot is not included because Hypothesis could not be installed here.